

Supply Chain Analysis: EDA Summary Report

PROJECT WORKFLOW

1. Data Cleaning: Handled missing values and date formats.
2. SLA Calculation: Defined breach as (Actual Days > Planned Days).
3. Carrier/Region Analysis: Compared performance across 8 carriers and 3 global regions.
4. Attribute Testing: Analyzed 'Priority' and 'Fragile' shipment performance.
5. Statistical Testing: Used Correlation Matrices to find relationships between variables.

KEY FINDINGS

- Global SLA Breach Rate: 47% (Systemic Issue).
- Average Delay: 3.7 days across the network.
- Consistency: All carriers and regions show nearly identical failure rates.
- Value Leakage: High-cost shipping does not guarantee on-time delivery.

RISK ANALYSIS

- Fragile Items: 60% breach rate (Highest Risk category).
- Priority Shipments: 44% breach rate (Tag is currently ineffective).
- Lead Times: Current planning is unrealistic for operational realities.

FINAL CONCLUSION

The 47% failure rate is a systemic planning problem, not a vendor-specific issue. To fix this, we must transition from manual planning to Predictive Machine Learning.